



Copper Price Risk & Volatility: A Quantitative Analysis Using R

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Introduction

Copper is one of the most widely used industrial metals. It is purchased and utilized in large quantities by several sectors including construction, automotive and energy. Therefore, movements in its price can reflect changes in global economic activity, supply and demand conditions, as well as investor expectations.

Over the past two decades, copper has experienced periods of substantial price appreciation as well as sharp declines and episodes of elevated volatility. The 2008 global financial crisis, the disruptions associated with the COVID-19 pandemic, and most recent changes in global commodity markets are examples of periods in which copper prices have experienced significant fluctuations.

This project examines the historical behavior of copper futures prices **from 2007 to September 4, 2026** using quantitative analysis in **R**. The objective is to analyze the size as well as the frequency of price movements, volatility changes through different periods and determine whether periods of high volatility are associated with higher returns.

Data & Methodology

Item	Description
Data source	Yahoo Finance
Instrument	Copper futures
Period	January 2007- September 4, 2026
Price measure	Daily closing price
Return measure	Daily arithmetic returns
Volatility measure	30-day rolling standard deviation
Extreme movements	Daily returns $\geq \pm 5\%$
Downside risk	Standard deviation of negative daily returns.

Copper Price & Return Behavior

Copper's return profile has undergone considerable changes across the 2007-2026 period, with several stress episodes, recovery, and renewed price appreciation.

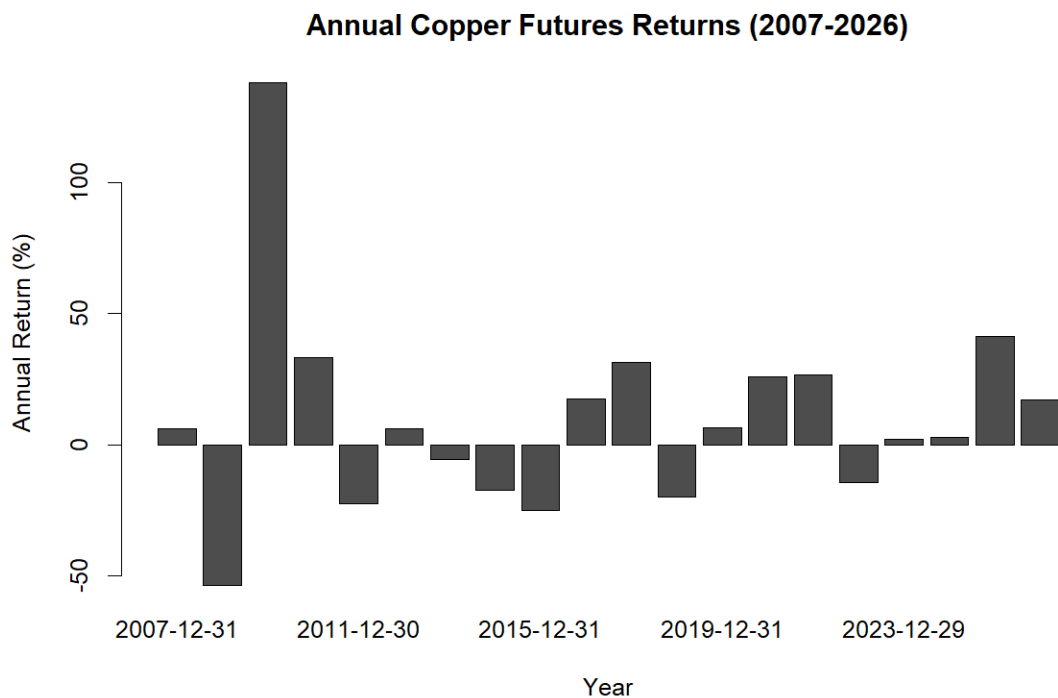
Copper experienced its most significant disruption during **2008–2009**. The metal recorded its weakest annual return in **2008 (–53.97%)**, along with the highest annual volatility in the sample (**2.99%**). Followed by an exceptional **138.53%** return in **2009**, it demonstrated how quickly performance can reverse following a major market shock. The decline in 2008 occurred during the global financial crisis while the magnitude of the subsequent rebound highlights the highly cyclical nature of copper prices.



During the **COVID-19 period** a different pattern emerged. Copper gained **25.77% in 2020** and **26.78% in 2021**, right before declining **14.58% in 2022**. This period is interesting because the initial pandemic disruption was followed by a strong recovery rather than a prolonged decline.

More recently, copper entered a relatively subdued period in **2023 (+1.97%) and 2024 (+2.72%)**, before recording a **41.24% return in 2025**. This stronger performance was accompanied by higher annual volatility of **2.38%**, compared with 1.42% in 2024. In **2026, copper has gained a further 17.18% year-to-date through September 4**, while annual volatility stands at 1.75%. This indicates that strong price performance during 2026 has not been accompanied by volatility at the same level as the most turbulent years in the sample.

Taken together, the results show that copper's return behavior is **highly dependent on the market environment**. The data move from the severe stress and sharp reversal of 2008–2009, through the pandemic recovery and subsequent correction, to a much stronger recent performance in 2025–2026. This makes copper particularly useful for examining how **returns and risk evolve across different market regimes**, rather than viewing its historical performance as a single uniform pattern.



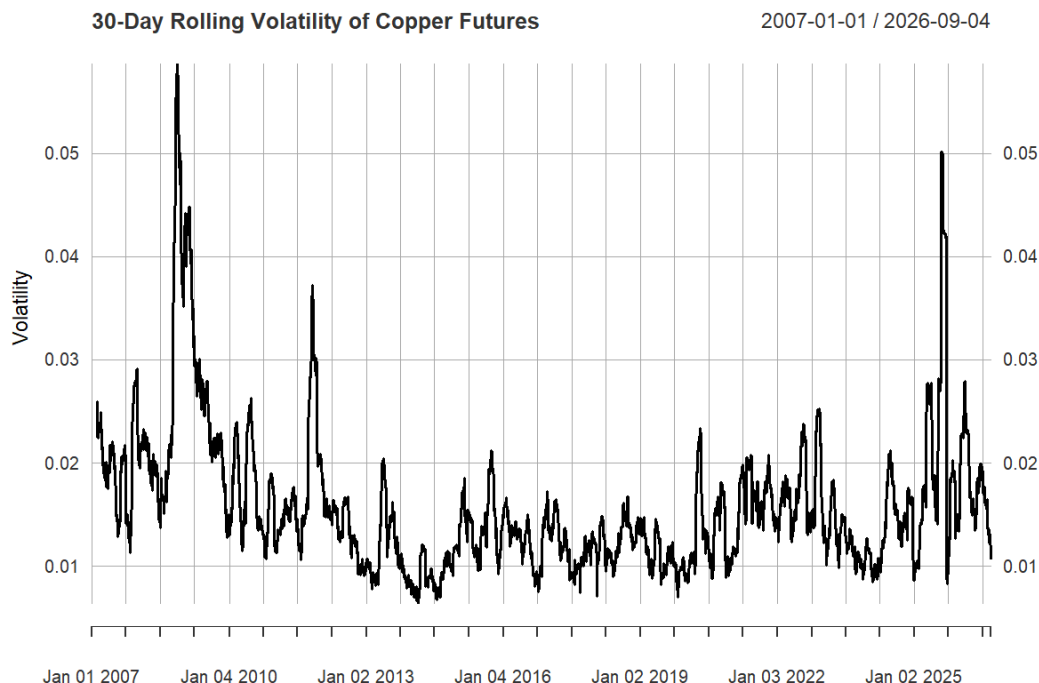
Volatility & Market Risk

The return analysis showed that copper's performance can change significantly across different market environments. The next question is whether **the level of price volatility changes in a similar way over time.**

Results pointed out that volatility is **time-varying**. Annual volatility ranged from **0.95% in 2014 to 2.99% in 2008**, while the 30-day rolling measure reached a maximum of **5.87% in November 2008**. Therefore, periods of elevated risk can be more intense compared to what copper's typical day-to-day behavior would suggest.

Elevated volatility was not limited to the GFC. The period of 2020–2022 also showed higher volatility than the relatively calm years preceding it, while **2025 recorded annual volatility of 2.38%**, substantially above **1.42% in 2024**. By September 4, 2026, the latest 30-day volatility had fallen to **1.07%**, suggesting that the recent increase in price activity has not translated into persistently elevated short-term volatility.

From a risk-management perspective there is a pattern that emerges. **Copper's risk is not constant. It expands and contracts over time.** Exposure to the same copper position can carry very different risk levels depending on the market environment.



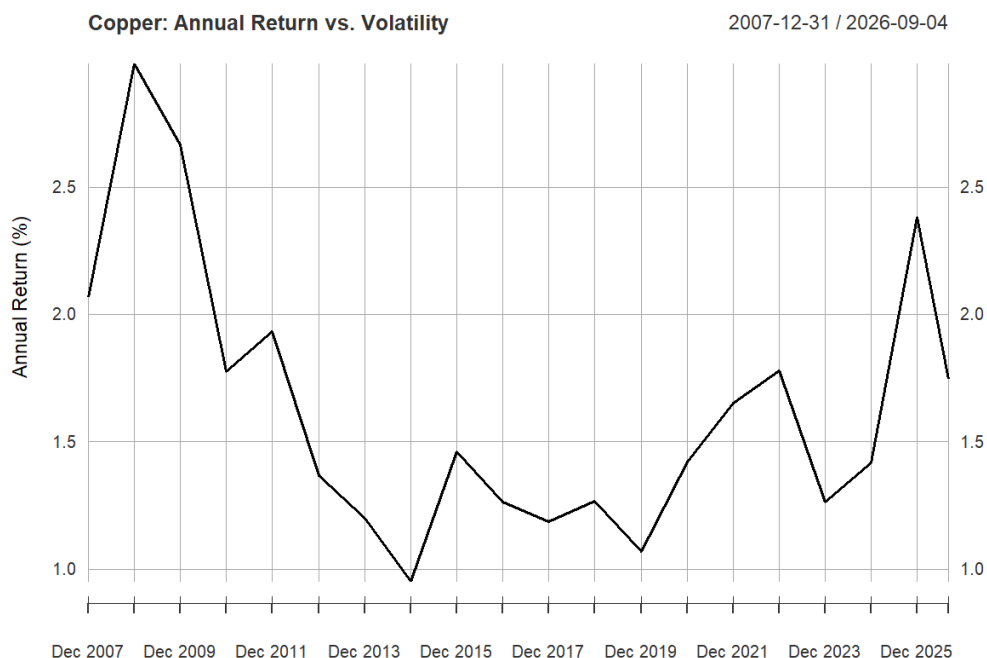
Return–Risk Relationship

After looking at how copper returns and volatility changed over time, the next question is whether **taking on more risk has historically been rewarded with higher returns**.

The analysis shows a **positive but weak correlation of 0.236 between annual returns and annual volatility**. This indicates that years with higher volatility tended to be associated with somewhat higher returns, but the relationship was far from consistent.

Recent data illustrate this well. In **2025**, copper delivered a **41.24% return** alongside relatively high annual volatility of **2.38%**. However, the relationship does not hold uniformly across the sample. For example, **2008 combined the highest annual volatility in the dataset (2.99%) with a -53.97% return**. This demonstrates that higher risk can accompany both substantial gains and substantial losses.

From an **investment perspective**, the distinction between realized volatility and expected return is important. Higher realized volatility should not automatically be interpreted as higher expected performance. For copper, periods of elevated risk can produce significant opportunities, but they can also result in substantial losses.



Key Findings & Interpretation

The analysis conducted shows that copper is not a commodity that should be viewed as an investment of simply high- or low-level risk. On the contrary, its historical behavior shows **substantial variation across market environments**, with both returns and volatility changing over time.

Three findings stand out:

1. **Copper's performance varied significantly over time.** It fell hard during the 2008–2009 crisis, rebounded strongly afterward, surged again in 2020–2021, and delivered strong returns in 2025–2026.
2. **Volatility has also fluctuated strongly.** Annual volatility ranged from 0.95% to 2.99%, and short-term volatility spiked much higher during periods of market stress.
3. **Returns and volatility are only weakly linked.** The correlation between annual returns and volatility is just 0.236, meaning higher risk hasn't reliably produced higher returns.

As a result, copper exposure cannot be assessed using a single historical risk estimate. For investors and firms with copper-related exposure, the **market environment and prevailing level of volatility** can materially affect the risk associated with a position. The analysis therefore suggests that **the key characteristic of copper-market risk is not simply its magnitude, but its variability over time.**

Limitations

Limitation	
Futures data	The analysis uses copper futures prices rather than spot prices, so futures-market characteristics may affect the results.
Historical analysis	The results describe past market behavior and should not be interpreted as forecasts of future returns or volatility.

Limited variables	The analysis focuses on price and return data and does not explicitly model factors such as demand, inventories, interest rates, or supply disruptions.
Data source	The historical series is obtained from Yahoo Finance and may differ from other market data providers due to methodology or contract treatment.

Conclusion

The above analysis showed that **copper-market risk is highly variable over time**. In addition, higher volatility has not consistently produced higher returns. This suggests that copper's risk profile cannot be assessed through a single measure or historical average. For financial analysts and market participants, the key takeaway is that **copper exposure needs to be evaluated in the context of the prevailing market environment**. Understanding how volatility changes over time can therefore be more informative for risk assessment than simply looking at historical returns.

This project demonstrates how **R can be used to transform historical market data into quantitative measures of return and price risk**, providing a practical framework for analyzing commodity-market behavior.